

# FishLodger

A Progressive Web App Up-to-Date Fish Market Monitoring System with Retailer Information, Supply Source Identification, Automated Report Generation, and Predictive Analytics for Lucena Fish Port Complex

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# Introduction

The Lucena Fish Port Complex is a developing regional fish trading center in Southern Luzon that serves as a vital economic hub in the marine supply chain. By receiving daily fish landings from coastal fleets and facilitating the distribution of marine products to various retailers and local markets, the fish port plays a crucial role in ensuring regional food security and supporting the livelihoods of local fishing communities.

# Objectives

The general objective of this study will be to enhance the operational efficiency and monitoring capabilities of the Lucena Fish Port Complex by developing a progressive web-based up-to-date fish market monitoring system with predictive analytics.

1. Identify the operational challenges and system requirements of the Lucena Fish Port Complex in terms of:

- a. Manual recording processes
- b. Retailer information management
- c. Supply source Identification
- d. Report generation and monitoring

2. Design and develop user registration, account management, and role-based access modules within FishLodger, specifically catering to:

- a. Retailer Account Registration and Management
- b. Supplier Account Registration and Management
- c. Consumer Access to Statistical Analysis, Predictive Insights, and Monitoring Dashboards through a View-Only Interface



**3.Design and develop core market monitoring modules to track day-to-day operations, focusing on:**

- a.Fish Arrival Monitoring**
- b.Supply Source Identification**
- c.Automated Report Generation**
- d.Dashboard and Data Visualization**

4.Design and develop data analytics features to provide insights and foresight for port operations, including:

a.Predictive Analytics

b.Market Trend Forecasts

5. Evaluate the developed system based on the ISO/IEC 25010  
Software Quality Standards in terms of:

- a. Functional Suitability
- b. Performance Efficiency
- c. Compatibility
- d. Usability
- e. Reliability
- f. Security



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# Scope and Limitation

FishLodger is a progressive web app designed to streamline operations at the Lucena Fish Port Complex through data management, supply tracing, report automation, and predictive analytics. Its limitations include an exclusive design for LFP, reliance on accurate manual data entry and vendor participation, vulnerability to internet outages, and predictive models that cannot account for sudden external disruptions like extreme weather.

# Methodology

the study details how the system is built to enhance data management, reporting, and operational transparency at the Lucena Fish Port Complex through price monitoring, supply tracking, and predictive analytics.

# Research Design

The researchers utilized a descriptive and developmental research design. This approach was selected to first accurately describe the existing manual price recording and supply monitoring processes at the Lucena Fish Port Complex (Descriptive), and subsequently design, build, and evaluate a digital solution in the form of a Progressive Web Application.

## Scrum Agile Methodology Phases Used:

- Sprint Planning
- Design
- Develop
- Test
- Daily Scrum
- Sprint Review
- Sprint Retrospective



# Data Gathering Procedure

Data was collected through administrator interviews, direct observations of manual processes, and document reviews of historical records. These methods allowed researchers to identify current operational challenges and gather the necessary data to build the system's database and AI forecasting model.

# Ethical Consideration

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# Wireframes



LUCENA FISH PORT COMPLEX PLATFORM

# FISHLODGER // PORT MONITORING

An integrated, offline-first Progressive Web Application (PWA) facilitating offline-enabled data entry, supply chain identification, AI price forecasting, and automated bulletin reporting.

ENTER COMMAND CENTER

OPERATOR LOGIN

LUCENA\_PORT\_COMMAND\_CENTER // LIVE\_FEED

|                                                                                                      |                                                                                       |                                                                                                      |
|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| <div>WEATHER ALERT</div> <div>29.4°C   Light Rain</div> <div>Wind: 3.2 m/s (Normal Operations)</div> | <div>LIVE ETA</div> <div>F/B Venture 422</div> <div>14m remaining (Tayabas Bay)</div> | <div>PREDICTIVE TREND</div> <div>Galunggong Forecast</div> <div>✓ Stable Price Trend (P210.00)</div> |
|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|

## Figure 1: Landing Page

## SYSTEM LOGIN

AUTHORIZED PERSONNEL ACCESS ONLY

USER IDENTIFIER

SECURITY KEY

LOGIN

NEW USER? [REGISTER IDENTITY](#)

Figure 2: Login

**FISHLODGER**

LOGINREGISTER

### CREATE IDENTITY

JOIN THE LUCENA FISH PORT DIGITAL ECOSYSTEM

FIRST NAMEJohn

LAST NAMEDoe

EMAIL ADDRESSjohn@example.com

CONTACT NUMBER09XX-XXX-XXXX

USERNAMEjohndoe123

PASSWORD\*\*\*\*\*

ACCOUNT ROLE

Port Retailer / Trader

**RETAILER STALL INFORMATION**

BUSINESS NAMEe.g. Maria's Fresh Fish

CHOOSE STALL UNIT-- SELECT UNIT --

BUSINESS ADDRESS

Enter complete stall or business address

INITIATE REGISTRATION

ALREADY REGISTERED? [LOGIN TO SESSION](#)

Figure 3: Register

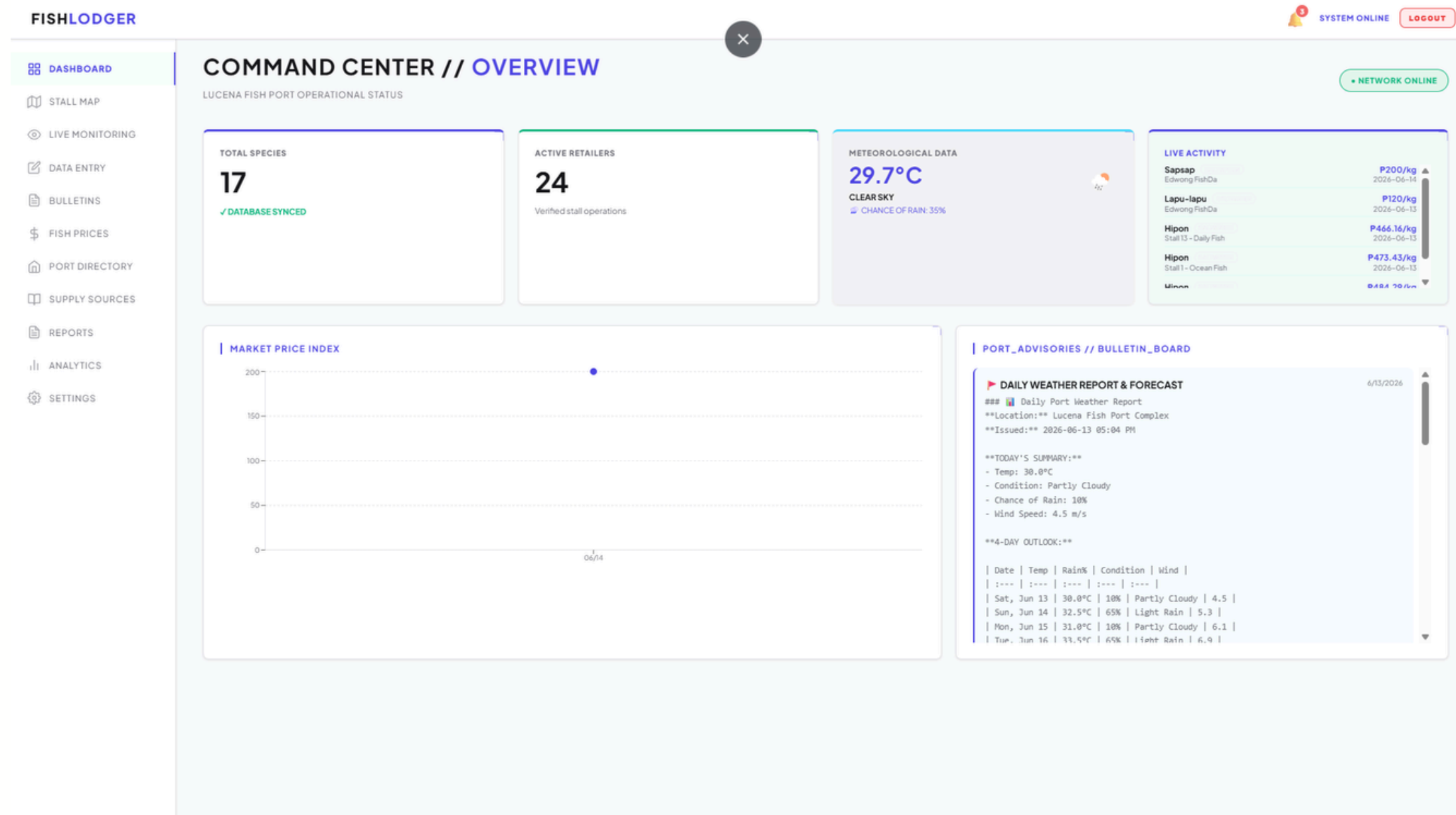


Figure 4: Dashboard

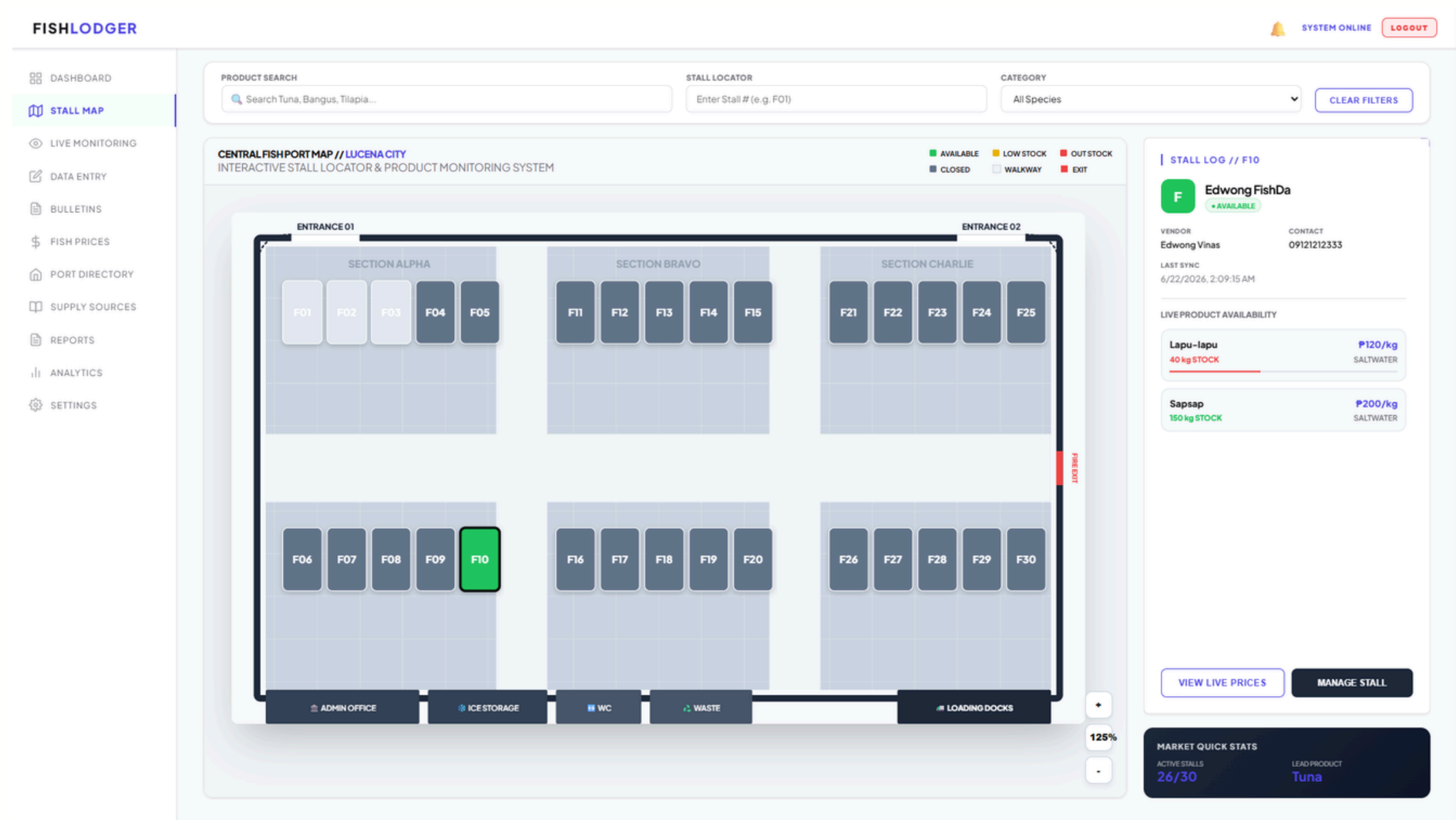


Figure 5: Stall Map



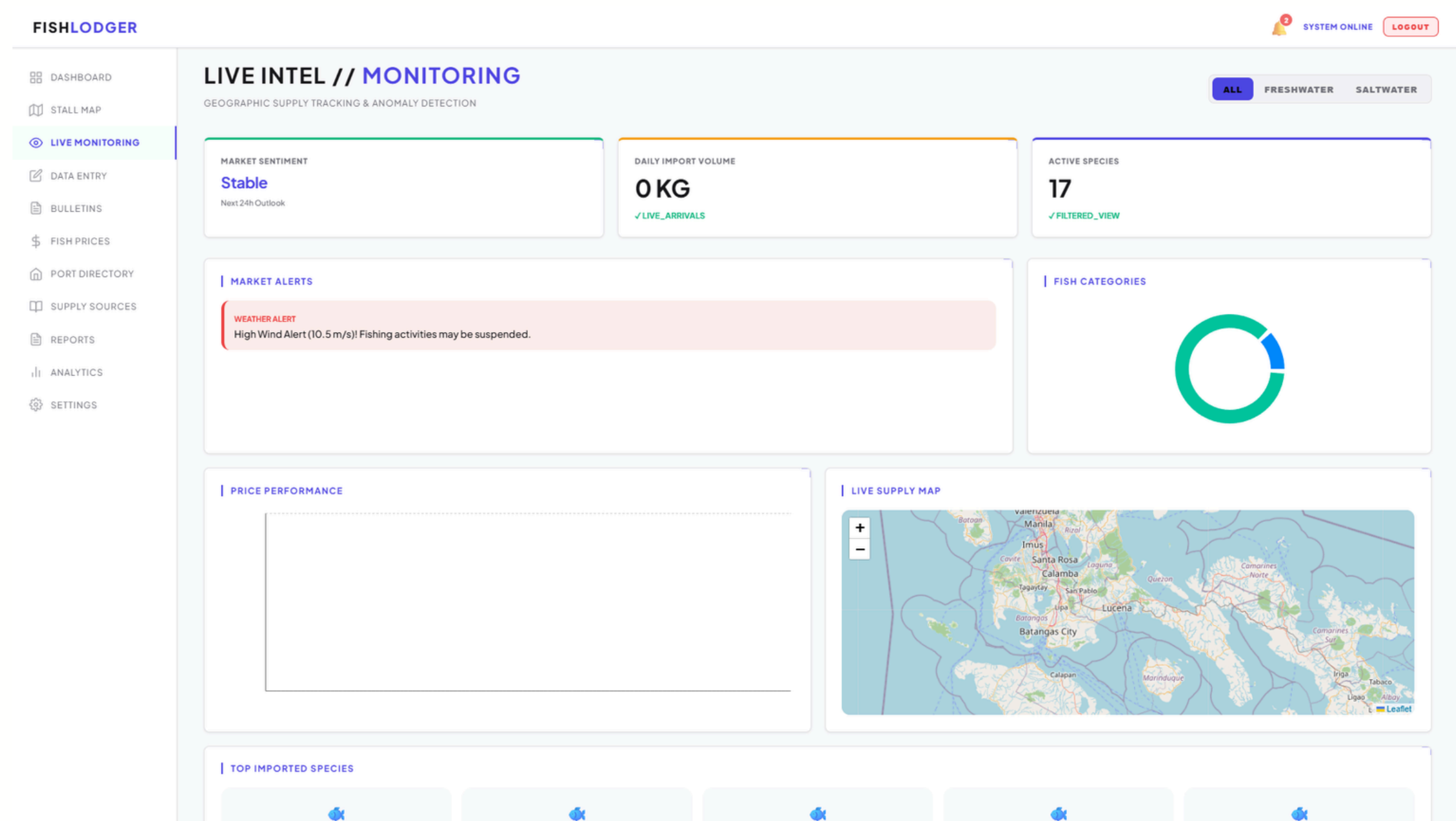


Figure 6: Live Monitoring

FISHLODGER

DASHBOARD

STALL MAP

LIVE MONITORING

DATA ENTRY

BULLETINS

FISH PRICES

PORT DIRECTORY

SUPPLY SOURCES

REPORTS

ANALYTICS

SETTINGS

SYSTEM ONLINE

LOGOUT

DATA ENTRY // OFFLINE ENABLED

RECORD DAILY MARKET OPERATIONS

RETAILER PRICE LOG

SUPPLIER DELIVERY LOG

FISH SPECIES

GALUNGGONG

BANGUS

TILAPIA

TUNA

TAMBAKOL

SAPSAP

+ ADD NEW SPECIES

Select Fish

RETAILER / STALL

STALL NUMBER

Search or type name...

ORIGIN OF FISH

Where was this fish caught?

PRICE (P/KG)

QUANTITY (KG)

SUBMIT TO SERVER

Figure 7: Data Entry



FISHLODGER

DASHBOARD

STALL MAP

LIVE MONITORING

DATA ENTRY

BULLETINS

FISH PRICES

PORT DIRECTORY

SUPPLY SOURCES

REPORTS

ANALYTICS

SETTINGS

BULLETIN MGMT // ADMIN PANEL

POST AND EDIT MARKET ADVISORIES

POST NEW BULLETIN

TITLE

e.g., Weather\_Warning: High Winds

CATEGORY

Information

CONTENT

POST BULLETIN

ACTIVE BULLETINS

Daily Weather Report & Forecast

6/13/2026, 5:04:16 PM | WEATHER

### 📄 Daily Port Weather Report

\*\*Location:\*\* Lucena Fish Port Complex

\*\*Issued:\*\* 2026-06-13 05:04 PM

\*\*TODAY'S SUMMARY:\*\*

- Temp: 30.0°C

- Condition: Partly Cloudy

- Chance of Rain: 10%

- Wind Speed: 4.5 m/s

\*\*4-DAY OUTLOOK:\*\*

[ Date | Temp | Rain% | Condition | Wind |

[ :-: | :-: | :-: | :-: | :-: ]

[ Sat, Jun 13 | 30.0°C | 10% | Partly Cloudy | 4.5 |

[ Sun, Jun 14 | 32.5°C | 65% | Light Rain | 5.3 |

[ Mon, Jun 15 | 31.0°C | 10% | Partly Cloudy | 6.1 |

[ Tue, Jun 16 | 33.5°C | 65% | Light Rain | 6.9 |

### 📄 Operational Guidance

- \*\*CAUTION:\*\* Unsettled conditions. Retailers should protect stock from rain/wind.

- \*\*NOTICE:\*\* Potential rainfall may affect fish drying operations. Ensure proper covering.

EDITDELETE

Figure 8: Bulletins



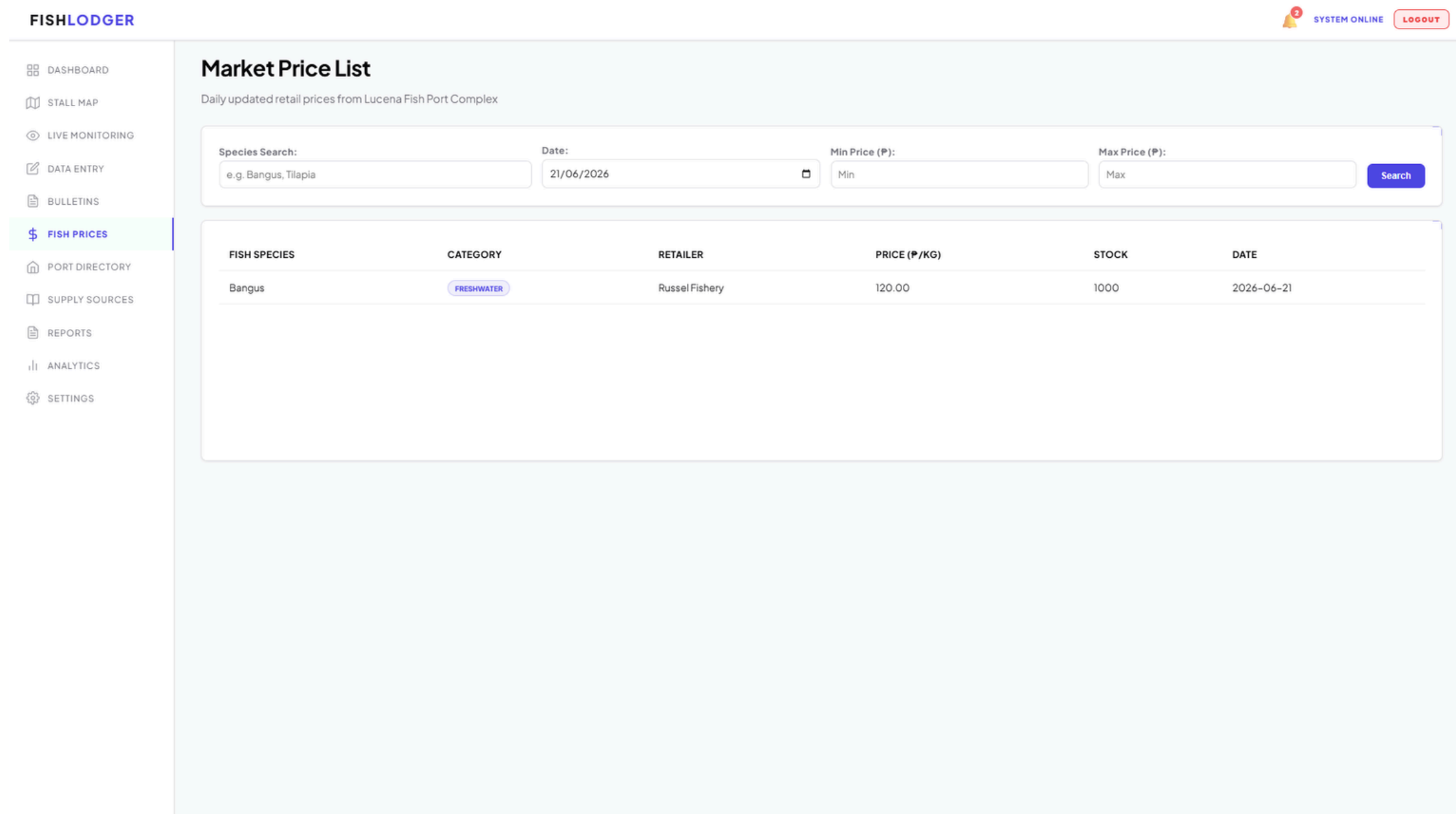


Figure 9: Fish Prices

FISHLODGER

DASHBOARD

STALL MAP

LIVE MONITORING

DATA ENTRY

BULLETINS

FISH PRICES

PORT DIRECTORY

SUPPLY SOURCES

REPORTS

ANALYTICS

SETTINGS

Port Directory

Verified registry of vessels and stall operators

SUPPLIERS (VESSELS)

RETAILERS (STALLS)

ACTIVE SUPPLY VESSELS

| VESSEL/BOAT NAME    | SUPPLIER NAME              | STATUS | CONTACT     |
|---------------------|----------------------------|--------|-------------|
| F/B Blue Wave 35    | Supplier Tayabas Bay 1     | DOCKED | 09177162877 |
| F/B Sea Spirit 44   | Supplier Tayabas Bay 2     | DOCKED | 09177474945 |
| F/B Morning Star 4  | Supplier Lamon Bay 1       | DOCKED | 09175506792 |
| F/B Sea Spirit 93   | Supplier Lamon Bay 2       | DOCKED | 09179297209 |
| F/B Morning Star 4  | Supplier Pagbilao Waters 1 | DOCKED | 09171981025 |
| F/B Morning Star 9  | Supplier Pagbilao Waters 2 | DOCKED | 09171465311 |
| F/B Morning Star 12 | Supplier Batangas Coast 1  | DOCKED | 09173440107 |
| F/B Sea Spirit 48   | Supplier Batangas Coast 2  | DOCKED | 09173063889 |
| F/B Venture 560     | Tayabas Bay Supplier 1     | DOCKED | 09123456789 |
| F/B Venture 304     | Tayabas Bay Supplier 2     | DOCKED | 09123456789 |
| F/B Venture 754     | Tayabas Bay Supplier 3     | DOCKED | 09123456789 |
| F/B Venture 533     | Lamon Bay Supplier 1       | DOCKED | 09123456789 |
| F/B Venture 782     | Lamon Bay Supplier 2       | DOCKED | 09123456789 |
| F/B Venture 209     | Lamon Bay Supplier 3       | DOCKED | 09123456789 |
| F/B Venture 990     | Pagbilao Waters Supplier 1 | DOCKED | 09123456789 |
| F/B Venture 858     | Pagbilao Waters Supplier 2 | DOCKED | 09123456789 |
| F/B Venture 260     | Pagbilao Waters Supplier 3 | DOCKED | 09123456789 |

SYSTEM ONLINE

LOGOUT

Figure 10: Port Directory

FISHLODGER

DASHBOARD

STALL MAP

LIVE MONITORING

DATA ENTRY

BULLETINS

FISH PRICES

PORT DIRECTORY

SUPPLY SOURCES

REPORTS

ANALYTICS

SETTINGS

2

SYSTEM ONLINE

LOGOUT

Supply Sources

Manage suppliers, vessels, and active fishing locations

ADD NEW SUPPLIER

Supplier Name \*

Boat Name \*

Contact Number \*

Fishing Location \*

Status \*

Arrival Date \*

Add Supplier

ACTIVE SUPPLIERS

| SUPPLIER NAME              | BOAT NAME           | STATUS | ARRIVAL DATE | ACTIONS |
|----------------------------|---------------------|--------|--------------|---------|
| Supplier Tayabas Bay 1     | F/B Blue Wave 35    | DOCKED | 2026-06-08   | Delete  |
| Supplier Tayabas Bay 2     | F/B Sea Spirit 44   | DOCKED | 2026-06-08   | Delete  |
| Supplier Lamon Bay 1       | F/B Morning Star 4  | DOCKED | 2026-06-08   | Delete  |
| Supplier Lamon Bay 2       | F/B Sea Spirit 93   | DOCKED | 2026-06-08   | Delete  |
| Supplier Pagbilao Waters 1 | F/B Morning Star 4  | DOCKED | 2026-06-08   | Delete  |
| Supplier Pagbilao Waters 2 | F/B Morning Star 9  | DOCKED | 2026-06-08   | Delete  |
| Supplier Batangas Coast 1  | F/B Morning Star 12 | DOCKED | 2026-06-08   | Delete  |
| Supplier Batangas Coast 2  | F/B Sea Spirit 48   | DOCKED | 2026-06-08   | Delete  |
| Tayabas Bay Supplier 1     | F/B Venture 560     | DOCKED | 2026-06-08   | Delete  |
| Tayabas Bay Supplier 2     | F/B Venture 304     | DOCKED | 2026-06-08   | Delete  |

Figure 11: Supply Sources

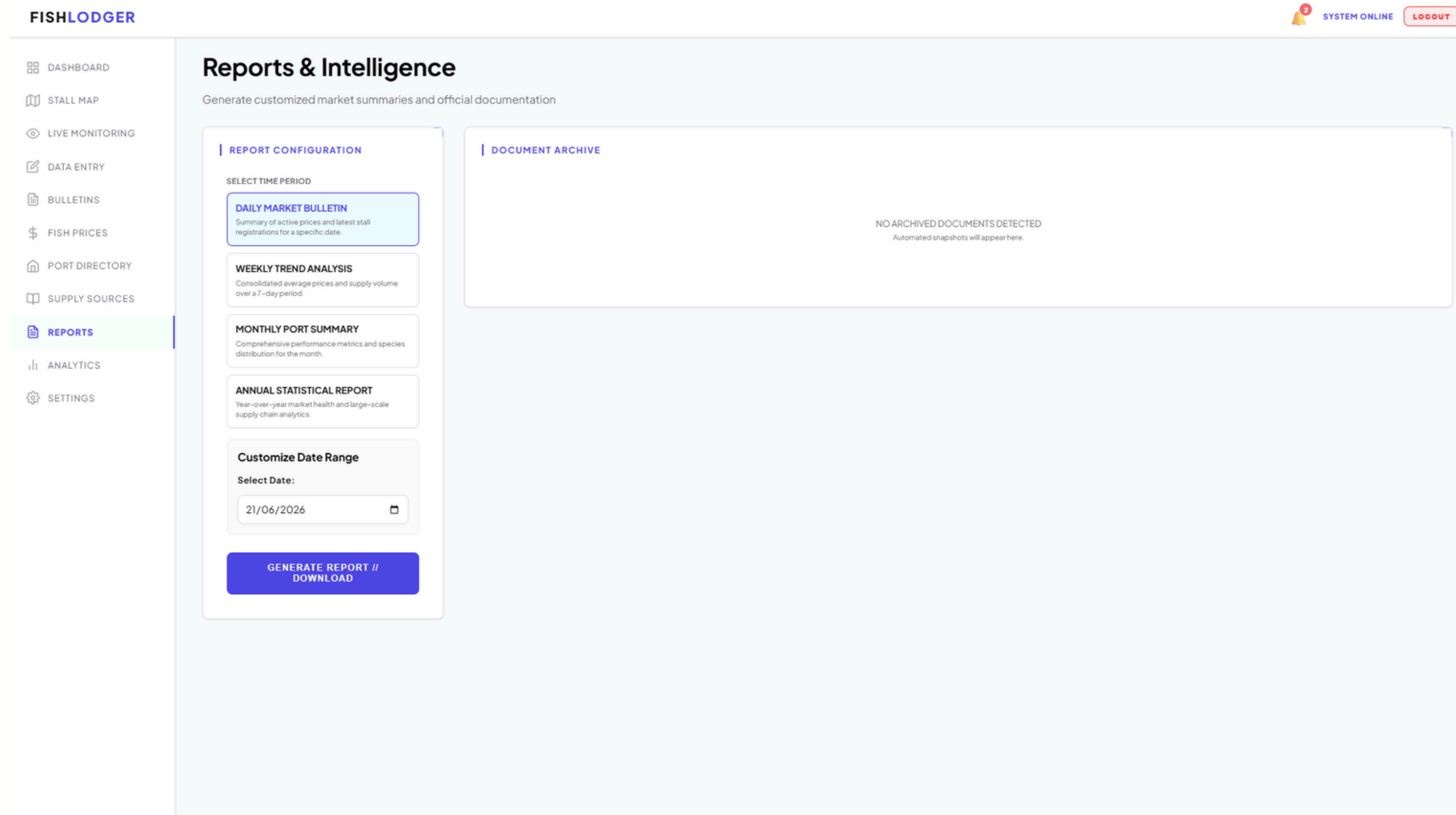


Figure 12: Reports

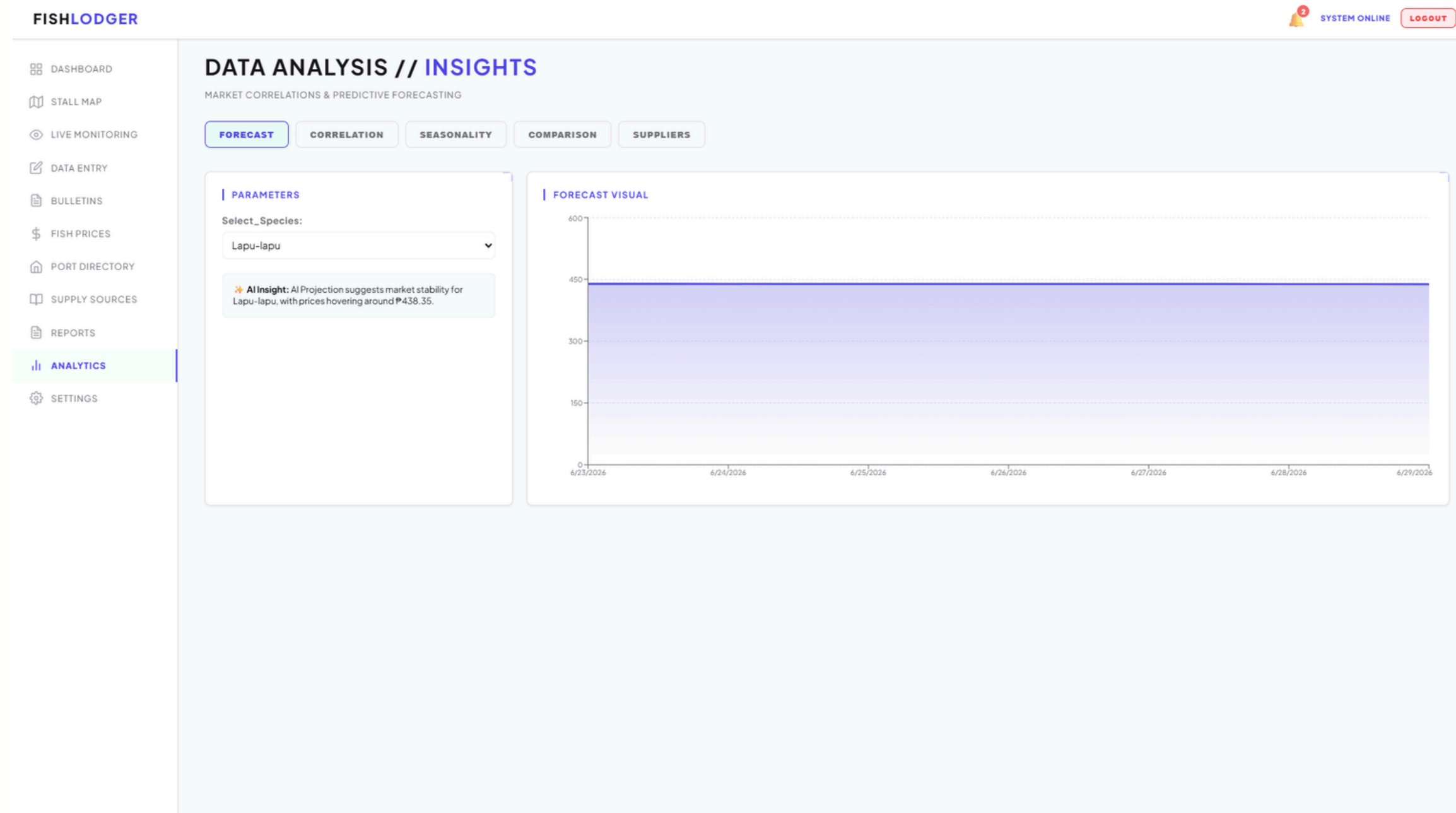


Figure 12: Analytics



FISHLODGER

DASHBOARD

STALL MAP

LIVE MONITORING

DATA ENTRY

BULLETINS

FISH PRICES

PORT DIRECTORY

SUPPLY SOURCES

REPORTS

ANALYTICS

SETTINGS

CONTROL CENTER // SETTINGS

PERSONAL PROFILE & SYSTEM CONFIGURATIONS

USER IDENTITY PROFILE

FIRST NAME

LAST NAME

EMAIL ADDRESS

admin@example.com

UPDATE\_PROFILE

SECURITY ENCRYPTION ACCESS

CURRENT PASSWORD

NEW PASSWORD

CONFIRM

RE-ENCRYPT\_PASSWORD

SYSTEM DIAGNOSTICS STORAGE

LOCAL CACHE SYNC STATUS

PENDING PRICELOGS

PENDING DELIVERY LOGS

0

0

APPLICATION METRICS

VERSION: 2.1.0-PRODUCTION

PROTOCOL: WEBSOCKET\_SECURE\_READY

STORAGE\_ENGINE: INDEXED\_DB\_DEXIE

Figure 13: Settings



Thank You